

Module 2 reflection

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The module covered a lot of basic practices like importing libraries, utilizing datasets, and displaying data via graphs. I have some basic experience with python, but I am still learning some of the names of libraries. This lab familiarized me with matplotlib, scikit, pandas, and NumPy. Briefly, scikit is imported for its utility in machine learning training, NumPy is used for efficient number processing in python, pandas helps with structuring the data, and matplotlib is used for visual representation of the data with graphs.

In the module, we used scikit to import one of the datasets (`load_iris`) for further analysis. The iris dataset contains information on 3 species, registering 4 features each as well as containing 50 different examples per species. Pandas was then used to structure this data for viewing specific details, giving us insight into the sorts of features that we can work with for future use. From there we were able to extract features that the iris has for each species. Specifically, they were averages of the sepal length, sepal width, petal length, and petal width as well as the 3 species used (setosa, versicolor, and virginica).

After identifying the features available to us, we then can begin plotting the information onto graphs using the matplotlib. We first use it to identify how many examples of each species are provided, which allows us to make sure there is an even and fair representation amongst the 3 species. Additionally, to visualize the variation in features of the examples provided of each species, we created a plot graph. In this instance, it shows us that setosa species on average have wider sepals while giving up length, while the other two species follow an opposite trend.

This module is short introduction into using a small dataset, but the fact that it is small helps in introducing concepts and utilizing them in tandem with others for analysis. Before this, I knew that several libraries often worked in tandem to efficiently work with data and code in machine learning, but I never quite understood the mechanics of how everything flowed. The biggest takeaway for me is how relatively simple it is to create graphs using the included data in order to make quick observations.